

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Vaishnavi.S (192521313) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

A multinational retail organization can implement a unified Big Data architecture to integrate customer data from online stores, mobile apps, POS systems, CRM, and loyalty programs.

Architecture:

Data Sources → Data Ingestion → Data Lake → Processing → Customer 360° → ML & Analytics → Recommendations

1. Data Ingestion & Storage: Collect data from online and offline channels and store it in a centralized Data Lake.
2. Big Data Processing: Use technologies such as Apache Spark for batch and real-time processing of large amounts of data.
3. Customer 360° & Analytics: Combine customer information to create a unified profile and analyze purchases, browsing history, and preferences.
4. Personalized Recommendations: Machine Learning models analyze customer behavior and provide real-time product recommendations and personalized offers.

Benefit: It eliminates fragmented data, provides real-time insights, improves customer experience, and increases sales.

2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Distributed Computing for Large-Scale Scientific Research:

Distributed computing is an approach where large scientific datasets and computational tasks are divided among multiple computers connected through a network. It is useful for scientific research involving simulations, experiments, and data analysis.

1. Parallel processing: The large workload is divided into smaller tasks and processed simultaneously by multiple computers.
2. Faster computation: Multiple processors work at the same time, reducing the overall processing time compared to a centralized system.
3. Scalability: More computers can be added when the amount of research data increases, allowing the system to handle larger workloads.
4. Better resource utilization: Distributed computing efficiently uses the processing power, memory, and storage of multiple machines.
5. Reliability: If one computer fails, other computers can continue processing, providing better fault tolerance.

Conclusion: Distributed computing with parallel processing provides faster computation, better resource utilization, and higher scalability than centralized systems for large-scale scientific research.

3.Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Fraud Detection Framework for a Digital Payment Platform:

A fraud detection framework uses Big Data analytics to identify suspicious transactions in real time and prevent financial fraud.

1. Data collection: Collect transaction data such as payment amount, location, device information, transaction time, and customer history.
2. Streaming data processing: Real-time technologies process transactions as they occur and immediately detect unusual activities, such as multiple transactions within a short period.
3. Data storage and processing: Store large volumes of historical transaction data in a Big Data platform and analyze it using distributed processing tools.
4. Predictive models: Machine learning models study previous fraudulent and genuine transactions to identify patterns and predict whether a new transaction is fraudulent.
5. Fraud prevention: When a transaction is predicted as suspicious, the system can generate an alert, block the transaction, or request additional verification.

Conclusion: Streaming processing provides fast, real-time fraud detection, while predictive models improve accuracy by identifying complex fraud patterns. Together, they help reduce fraud and improve the security of digital payments.

4.Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Challenges and Technologies for Multimedia Big Data:

Multimedia data such as videos, audio files, and images are large, complex, and produced continuously. Managing this data efficiently is a major challenge in a Big Data environment.

1. Storage challenges: Multimedia files require huge storage capacity and may have different formats, resolutions, and sizes.
2. Processing challenges: Processing large video, audio, and image files requires high computing power and takes more time than processing normal structured data.
3. Suitable storage: Distributed storage systems such as HDFS and cloud object storage can store large multimedia datasets across multiple servers.

4. Suitable processing: Apache Spark can be used for distributed processing and analysis of large multimedia datasets. GPUs can also speed up image and video processing.
5. Efficient management: Data compression, metadata management, partitioning, and replication can reduce storage requirements and improve data access and reliability.

Conclusion: Using distributed storage, cloud storage, parallel processing, and compression provides efficient and scalable management of multimedia Big Data.

5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

A cloud-based Big Data platform can use the following disaster recovery strategy:

Cloud Platform → Backup → Replication → Monitoring → Automated Failover → Recovery

1. Backup: Regularly back up important data and system configurations to secure cloud storage.
2. Redundancy: Use multiple servers and storage systems so that failure of one component does not stop the service.
3. Replication: Maintain copies of data in another server or geographical region to prevent data loss.
4. Automated failover: When the primary system fails, automatically switch to the backup system to reduce downtime.
5. Recovery: After the failure, restore the primary system and verify data consistency before returning to normal operation.

Thus, redundancy, replication, and automated failover provide high availability, protect data, reduce downtime, and ensure business continuity.

6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Poor data quality can reduce the accuracy and reliability of a large-scale analytics platform.

1. Inconsistency: Different formats or values for the same data can produce incorrect analysis and misleading results.
2. Duplication: Repeated records increase storage usage and may cause incorrect calculations and customer counts.

3. Missing values: Incomplete data can reduce the accuracy of reports, predictions, and machine learning models.
4. Data governance: Apply data validation, standard formats, deduplication, and suitable methods for handling missing values.
5. Monitoring and policies: Establish data quality rules, assign data ownership, and continuously monitor data for errors.

Thus, proper data governance improves data accuracy, consistency, and completeness, leading to more reliable analytics and better decision-making.

7.Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

Scalable Recommendation System for an Online Streaming Service:

Architecture:

User Activity → Data Collection → Data Lake → Spark Processing → ML Recommendation Model → Personalized Content

1. Data collection: Collect user data such as watch history, searches, ratings, likes, skips, and viewing time.
2. Distributed processing: Use Hadoop or Apache Spark to process large amounts of user data across multiple machines.
3. User behavior analytics: Analyze user preferences and viewing patterns to identify the type of content each user likes.
4. Recommendation model: Machine learning models generate personalized recommendations based on user behavior and similar users.
5. Example: If a user frequently watches action movies, the system analyzes this behavior and recommends similar action movies or series.

Thus, distributed processing improves scalability and speed, while user behavior analytics improves recommendation accuracy and increases user engagement.

8.Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Role of Cloud Computing in Modern Big Data Ecosystems:

Architecture:

Big Data Sources → Cloud Storage → Cloud Processing → Big Data Analytics → Results

Cloud computing provides on-demand storage and computing resources for managing large volumes of Big Data.

1. Scalability: Cloud resources can be increased or decreased according to the data workload, making it suitable for growing Big Data applications.
2. Cost management: Organizations pay mainly for the resources they use, reducing the need for expensive hardware. However, continuous or high usage can increase cloud costs.
3. Flexibility: Cloud platforms provide different storage, processing, and analytics services that can be quickly configured based on requirements.
4. Security: Cloud providers offer encryption, access control, monitoring, and backup facilities. However, data privacy, misconfiguration, and unauthorized access remain security concerns.
5. Overall evaluation: Cloud computing provides scalable, flexible, and cost-effective Big Data management, but organizations must carefully control costs and implement strong security measures.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Architecture:

GPS/IoT Sensors + Traffic Data + Vehicle Data → Data Ingestion → Big Data Processing → Real-Time Analytics → Route Optimization → Decision Making

1. Data collection: Collect real-time data from GPS devices, IoT sensors, traffic systems, weather information, and vehicle fuel sensors.
2. Real-time processing: Use technologies such as Apache Kafka and Spark Streaming to process vehicle and traffic data continuously.
3. Route optimization: Analyze traffic, distance, and road conditions to identify the fastest and shortest routes.
4. Fuel efficiency: Monitor vehicle speed, distance, traffic, and fuel consumption to reduce unnecessary fuel usage and idle time.
5. Operational decisions: Managers can use real-time dashboards and alerts to respond quickly to traffic, delays, vehicle problems, and delivery changes.

Example: If heavy traffic is detected on a delivery route, the system immediately suggests an alternative route, reducing travel time and fuel consumption.

Thus, Big Data technologies improve route planning, fuel efficiency, delivery performance, and real-time decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Ethical and Legal Challenges in Big Data:

Big Data collection and analytics can create ethical and legal issues related to privacy, security, and responsible use of personal information.

Architecture:

Data Collection → Data Storage → Data Analytics → Privacy & Security Controls → Responsible Data Usage

1. Privacy: Collecting personal data without proper consent can violate user privacy and create ethical concerns.
2. Security: Data breaches, unauthorized access, and misuse of sensitive information can cause financial and personal harm.
3. Bias and fairness: Poor-quality or biased data can produce unfair results in analytics and decision-making.
4. Organizational policies: Organizations should obtain user consent, collect only necessary data, use encryption and access controls, and regularly audit data usage.
5. Legal compliance and trust: Follow data protection laws, provide transparency about data collection, allow users to control their data, and maintain proper data retention policies.

Thus, strong privacy, security, transparency, and compliance policies ensure responsible Big Data usage and improve user trust.

